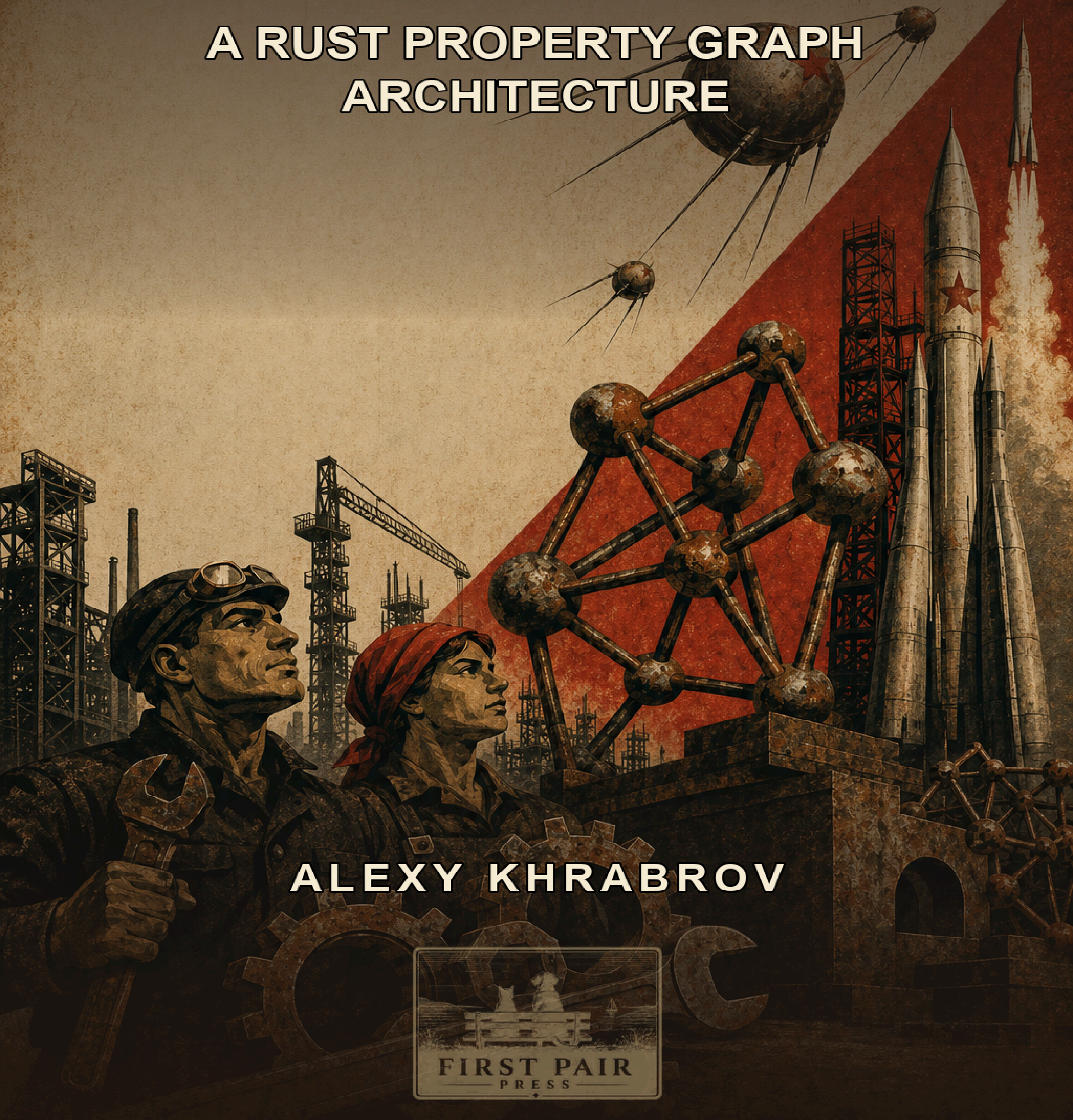
GRUST

A RUST PROPERTY GRAPH ARCHITECTURE



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Grust

A Rust Property Graph Architecture

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Contents

1 Preface

Grust is a small Rust property graph API with a large architectural promise: application code should be able to build, validate, traverse, and persist graph data without committing itself to a database query language too early.

That goal matters because many graph-shaped Rust applications live between worlds. A crawler wants a local graph while it extracts facts. An indexing pipeline wants deterministic target state. A data product may begin with tests and an in-memory backend, then later need SurrealDB, FalkorDB, PostgreSQL, LanceDB, Sail, or another backend. Grust gives those applications one model to program against:

```
Graph = nodes + edges
Node  = id + label + properties
Edge  = optional id + from + to + label + properties
```

This book is a guided tour of that model. It covers the workspace architecture, the Rust concepts the code leans on, the core graph types, traversal IR, backend contracts, implemented backend profiles, query safety, semantic-model projection, and practical examples.

2 1. The Shape of Grust

Grust is not trying to be every graph library for Rust. It is not an algorithm crate like `petgraph`, and it is not a thin wrapper around a single database. It is a property graph layer: a compact model of labeled nodes, labeled edges, and typed properties that can be carried across storage engines.

The current workspace is arranged around one core crate, one public facade, and backend or integration crates:

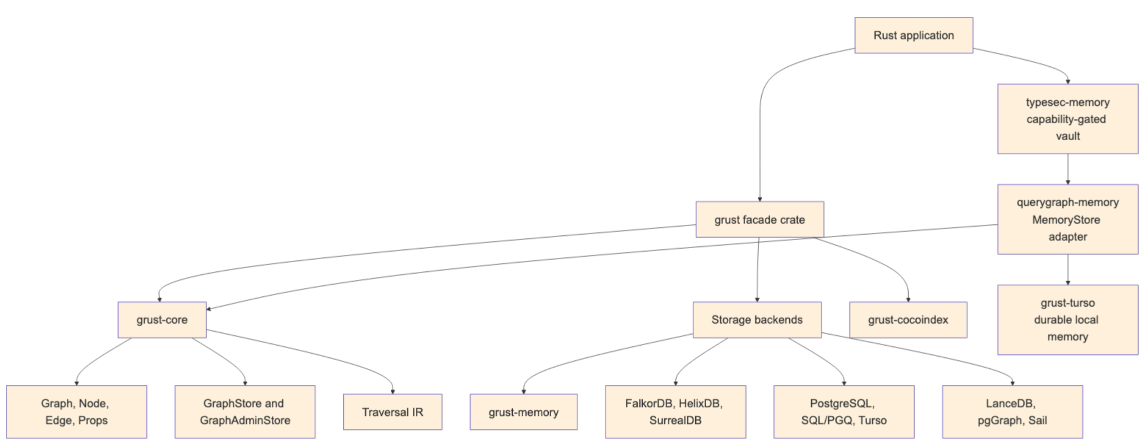

Figure 1: Diagram 1

The `grust-core` crate owns the durable concepts. It defines identifiers, labels, property values, graph builders, schemas, traversals, error types, load reports, and backend traits. The `grust` crate re-exports those pieces and gates backend crates behind Cargo features.

Core also owns the small lowering helpers that have to mean the same thing everywhere. `relationship_type` normalizes edge labels for backends that need database-safe relationship names. `schema_identifier` normalizes schema labels and fields for SQL-like typed surfaces. `edge_key` gives table and export backends the same structural fallback key for an edge when the caller has not provided an explicit `EdgeId`.

That split is important. Core model code stays light, deterministic, and mostly dependency-free. Backends are allowed to depend on Redis, SurrealDB, LanceDB, PostgreSQL, Spark Connect, HTTP clients, gRPC clients, or SDKs without dragging those dependencies into every Grust user.

3 2. The Core Property Graph

At the center of Grust are three public data structures:

```
pub struct Graph {
    pub nodes: Vec<Node>,
    pub edges: Vec<Edge>,
}

pub struct Node {
    pub id: NodeId,
    pub label: Label,
    pub props: Props,
}

pub struct Edge {
    pub id: Option<EdgeId>,
    pub from: NodeId,
    pub to: NodeId,
    pub label: Label,
    pub props: Props,
}
```

`NodeId`, `EdgeId`, and `Label` are newtypes over `String`. They are small wrappers, but they carry meaning through Rust's type system. A function that expects a `Label` cannot accidentally receive an `EdgeId` without an explicit conversion. The wrappers implement common conversion traits such as `From<&str>` and `From<String>`, so the public API remains ergonomic.

Properties are stored in a `BTreeMap<String, Value>`:

```
pub enum Value {
    Null,
    Bool(bool),
    Int(i64),
    Float(f64),
    String(String),
```

```
    DateTime(RfcDate),
    Decimal(Decimal),
    Duration(Duration),
    StringArray(Vec<String>),
    IntArray(Vec<i64>),
    FloatArray(Vec<f64>),
    Path(PathValue),
    Graph(GraphValue),
    Json(serde_json::Value),
}
```

The use of `BTreeMap` is a quiet but useful choice. Iteration order is stable, which makes generated JSON, tests, and backend query generation more predictable. That is especially helpful in a project whose backends turn one graph into SQL, Cypher-like queries, Arrow batches, JSON target state, or SDK calls.

`GraphValue` deduplicates relationships by explicit ID or structural identity. Those identities use length-framed components rather than payload delimiters, so punctuation and control text inside an ID, label, or endpoint cannot make two distinct relationship values look identical. Persisted compatibility keys have a different contract: `checked_edge_key` rejects U+001F in every component before a tabular or export backend materializes the delimiter-based key.

`Node::new` also inserts an `id` property if one is not already present. That means a Grust node has a first-class typed identity and an ordinary property view of that identity, which helps backends that identify records through property maps.

4 3. Building Graphs

Most application code should not construct every `Node` and `Edge` by hand. It should use `GraphBuilder`:

```
use grust::prelude::*;

let mut builder = GraphBuilder::new();

let talk = builder
    .node("Talk", "talk:rust-graph-api")
    .prop("title", "A Modern Graph API for Rust")
    .prop("year", 2026_i64)
    .finish();

let speaker = builder
    .node("Person", "person:ada")
    .prop("name", "Ada Example")
    .finish();

builder
    .edge("PRESENTED_BY", &talk, &speaker)
    .prop("source", "conference-schedule")
    .finish();
```

```rust
let graph = builder.build();
```

The builder deduplicates nodes by `NodeId`. If an existing node has the same label, additional properties are merged into it. Edges default to `EdgePolicy::DedupeByFromLabelTo`, so the tuple `(from, label, to)` is treated as the relationship identity. Domains that need true multi-edges can opt into `EdgePolicy::AllowDuplicates`.

```rust
let mut builder = GraphBuilder::new()
    .edge_policy(EdgePolicy::AllowDuplicates);
```

This is an example of Grust's overall style: put the common property-graph case on the simple path, but leave an explicit escape hatch for graph models with different identity rules.

4.1 Loading and Saving Graph Documents

A `Graph` can also move through textual document formats without touching a backend. The core crate exposes paired constructors and serializers for YAML, JSON, and XML:

```rust
let graph = Graph::from_yaml(yaml_text)?;
let yaml_text = graph.to_yaml()?;

let graph = Graph::from_json(json_text)?;
let json_text = graph.to_json()?;

let graph = Graph::from_xml(xml_text)?;
let xml_text = graph.to_xml()?;
```

These methods are useful for fixtures, migration inputs, examples, audits, and small graph interchange files. They all feed the same validation path, so a document with duplicate node ids or an edge that points at a missing node fails before it reaches a store.

YAML and JSON share the same document shape. A property value can be written as a plain JSON-like scalar when the type is obvious, or as the tagged Grust `Value` representation when the exact variant matters:

```json
{
  "nodes": [
    {
      "id": "talk:rust-graph-api",
      "label": "Talk",
      "props": {
        "title": "A Modern Graph API for Rust",
        "year": 2026,
        "tracks": {
          "type": "string_array",
          "value": ["rust", "graphs"]
        }
      }
    },
    {
```

```json
      "id": "person:ada",
      "label": "Person",
      "props": {
        "name": "Ada Example"
      }
    }
  ],
  "edges": [
    {
      "label": "PRESENTED_BY",
      "from": "talk:rust-graph-api",
      "to": "person:ada",
      "props": {
        "source": "conference-schedule"
      }
    }
  ]
}
```

The XML form is more explicit because XML has no native object or array type. Properties are represented as repeated `prop` entries with a `key` and a tagged `value`:

```xml
<graph>
  <nodes>
    <node>
      <id>talk:rust-graph-api</id>
      <label>Talk</label>
      <props>
        <prop>
          <key>title</key>
          <value>
            <type>string</type>
            <value>A Modern Graph API for Rust</value>
          </value>
        </prop>
      </props>
    </node>
    <node>
      <id>person:ada</id>
      <label>Person</label>
    </node>
  </nodes>
  <edges>
    <edge>
      <label>PRESENTED_BY</label>
      <from>talk:rust-graph-api</from>
      <to>person:ada</to>
    </edge>
  </edges>
</graph>
```

Serialization removes the generated `id` property from node property maps when it only mirrors the node's `NodeId`. That keeps exported documents readable without changing what `Node::new` guarantees after loading.

5 4. Typed Graph Ingestion with garde and zod-rs

The core `Graph`, `Node`, and `Edge` types are intentionally dynamic. A node has a label and a property map; an edge has a label, endpoints, and a property map. That dynamic shape is the right interchange format for backends, graph documents, audits, and cross-system movement. Application code, however, often starts with typed Rust structs: a `Person`, a `Project`, a `WorksOn` relationship, or some domain-specific event. Grust's typed ingestion layer connects those two worlds.

The typed layer is optional. It is enabled through Cargo features:

```
[dependencies]
grust = { package = "grust-graph", version = "0.13.2", features = ["typed-
garde"] }
```

`typed-garde` adds Rust-struct validation and typed lowering. A second feature, `typed-zod-rs`, layers raw JSON shape validation on top:

```
[dependencies]
grust = { package = "grust-graph", version = "0.13.2", features = ["typed-zod-
rs"] }
```

`typed-zod-rs` implies `typed-garde`. That relationship matters: zod-rs checks untrusted JSON shape first, Serde turns the JSON into Rust values, and garde checks typed domain invariants before Grust lowers the value into a normal `Node` or `Edge`.

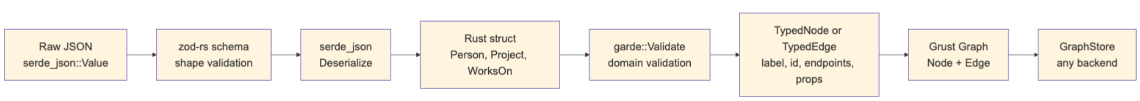

Figure 2: Diagram 2

The important promise is that typed ingestion does not create a second graph model. Once validation succeeds, everything becomes the ordinary Grust graph shape. Backends do not need to know whether a node came from `GraphBuilder`, a YAML document, a typed Rust struct, or a zod-rs-validated JSON payload.

5.1 garde: Typed Rust Validation

`garde` is a Rust validation library built around a derive macro. You attach validation rules to fields, derive `garde::Validate`, and then call `validate()` or `validate_with(...)`. The rules live near the Rust type, so they travel with the domain model instead of being hidden in a loader function.

Common garde rules include:

- `length(min = 1)` for nonempty strings and collections.
- `range(min = 1, max = 100)` for numeric bounds.
- `inner(...)` for validating items inside a collection.
- `custom(...)` for domain-specific validation functions.

- `dive` for validating nested values that also implement `Validate`.

In Grust, a typed node implements `TypedNode`. It supplies a graph label and a stable node id. The default property conversion serializes the struct through Serde and converts the resulting object fields into Grust properties:

```
use grust::prelude::*;
use grust::typed::garde;
use serde::Serialize;

#[derive(Debug, Serialize, garde::Validate)]
#[garde(allow_unvalidated)]
struct Person {
    #[garde(length(min = 1))]
    id: String,
    #[garde(length(min = 1))]
    name: String,
    #[garde(length(min = 1), inner(length(min = 1)))]
    skills: Vec<String>,
}

impl TypedNode for Person {
    const LABEL: &'static str = "Person";

    fn node_id(&self) -> NodeId {
        format!("person:{}", self.id).into()
    }
}
```

Typed edges implement `TypedEdge`. They provide a label and endpoint ids:

```
#[derive(Debug, Serialize, garde::Validate)]
#[garde(allow_unvalidated)]
struct WorksOn {
    #[garde(length(min = 1))]
    person_id: String,
    #[garde(length(min = 1))]
    project_id: String,
    #[garde(range(min = 1, max = 100))]
    allocation_percent: u8,
}

impl TypedEdge for WorksOn {
    const LABEL: &'static str = "WORKS_ON";

    fn source_node_id(&self) -> NodeId {
        format!("person:{}", self.person_id).into()
    }

    fn target_node_id(&self) -> NodeId {
```

```rust
        format!("project:{}", self.project_id).into()
    }
}
```

`TypedGraphBuilder` validates and lowers these values:

```rust
let mut builder = TypedGraphBuilder::new();

builder.add_node(&Person {
    id: "nia".to_string(),
    name: "Nia".to_string(),
    skills: vec!["rust".to_string(), "graphs".to_string()],
})?;

builder.add_edge(&WorksOn {
    person_id: "nia".to_string(),
    project_id: "grust".to_string(),
    allocation_percent: 80,
})?;

let graph = builder.build();
```

Validation fails before graph construction. If `allocation_percent` is `0`, the `range(min = 1, max = 100)` rule produces a Grust schema error and the edge is not added. This keeps invalid domain facts out of the graph rather than relying on a backend to reject them later.

Typed values can also be reconstructed from ordinary Grust reads. The read path decodes node or edge properties back through serde, runs garde validation, and checks that the typed identity still matches the graph value:

```rust
let stored = store
    .get_node(&NodeId::new("person:nia"))
    .await?
    .expect("person exists");

let person = Person::from_node(&stored)?;
assert_eq!(person.id, "nia");
```

Edges follow the same pattern with `WorksOn::from_edge(&edge)?`. For validation contexts, use `from_node_with` or `from_edge_with`.

5.2 Typed and Untyped Graphs Coexist

The typed layer is an ingestion layer over the ordinary `GraphBuilder`. It does not force an all-or-nothing choice. A graph can start as a document, accept raw nodes and edges, and then be extended with typed values:

```rust
let existing = Graph::new(
    vec![Node::new("Document", "doc:garde-proposal", Props::new())],
    Vec::new(),
);
```

```rust
let mut builder = TypedGraphBuilder::from_graph(existing);

builder.add_node(&Person {
    id: "nia".to_string(),
    name: "Nia".to_string(),
    skills: vec!["rust".to_string(), "graphs".to_string()],
})?;

builder.add_raw_edge(Edge::new(
    "AUTHORED",
    "person:nia",
    "doc:garde-proposal",
    Props::new(),
));

let graph = builder.build();
```

The coexistence API is explicit:

- `TypedGraphBuilder::from_graph(graph)` starts from an existing `Graph`.
- `TypedGraphBuilder::from_builder(builder)` starts from an existing `GraphBuilder`.
- `add_raw_node(node)` and `add_raw_edge(edge)` add ordinary Grust values.
- `into_builder()` returns the inner `GraphBuilder` when lower-level builder operations are needed.

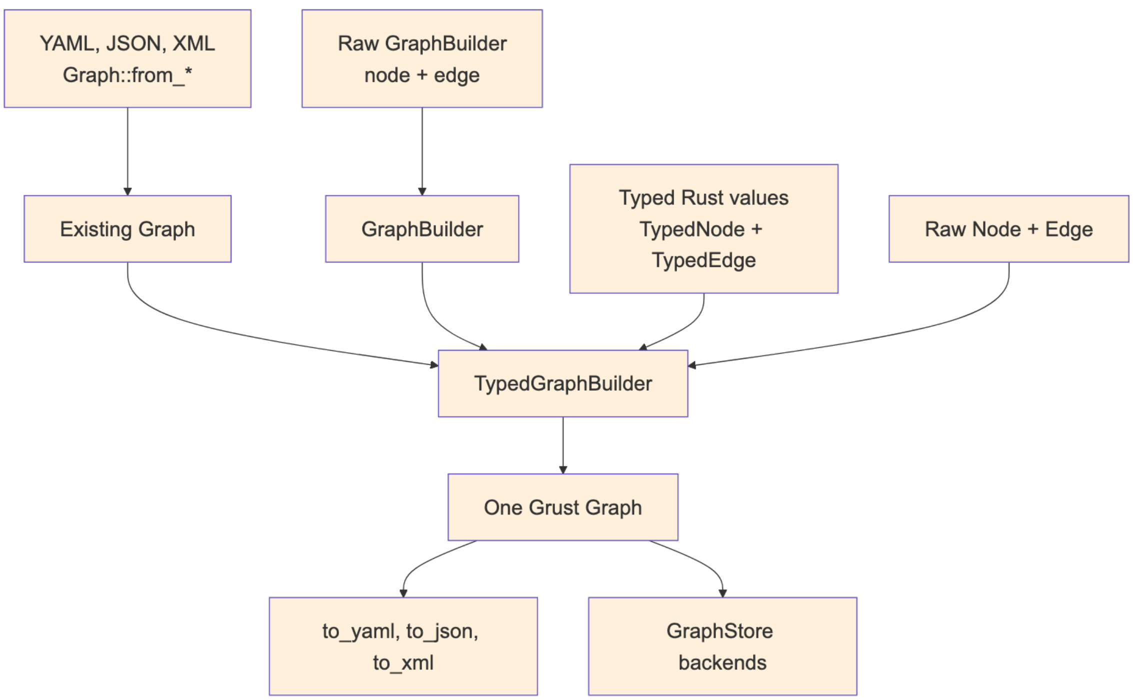

Figure 3: Diagram 3

This is useful during migrations. A project can keep loading existing graph documents while adding typed definitions for the domain facts that benefit most from Rust validation. Over time, more labels can move to typed constructors without breaking the storage or document formats.

5.3 What Zod Means Here

Zod is best known from TypeScript. It lets developers define a runtime schema and validate untrusted input before treating it as application data. In TypeScript, this closes an important gap: static types disappear at runtime, so a value received from JSON, an HTTP request, a form, or a message queue still needs runtime validation.

Rust has a different type system, but the boundary problem still exists. External data arrives as bytes or JSON. Serde can deserialize those bytes into a Rust struct, but it is often useful to separate two questions:

1. Does this JSON have the expected shape?
2. Does the resulting Rust value satisfy my domain rules?

`zod-rs` answers the first question. `garde` answers the second. Grust's `typed-zod-rs` feature wires those stages together:

```rust
use grust::prelude::*;
use serde::{Deserialize, Serialize};
use serde_json::json;
use zod_rs::prelude::{Schema, number, object, string};

#[derive(Debug, Deserialize, Serialize, garde::Validate)]
#[garde(allow_unvalidated)]
struct Person {
    #[garde(length(min = 1))]
    id: String,
    #[garde(length(min = 1))]
    name: String,
    #[garde(length(min = 1), inner(length(min = 1)))]
    skills: Vec<String>,
}

impl TypedNode for Person {
    const LABEL: &'static str = "Person";

    fn node_id(&self) -> NodeId {
        format!("person:{}", self.id).into()
    }
}

let person_schema = object()
    .field("id", string().min(1))
    .field("name", string().min(1))
    .field("skills", string().min(1).array())
    .strict();
```

```
let json = json!({
    "id": "nia",
    "name": "Nia",
    "skills": ["rust", "graphs"]
});

let person: Person = parse_typed_json(&person_schema, &json)?;
```

The same schema can feed the graph builder directly:

```
let mut builder = TypedGraphBuilder::new();
builder.add_node_from_json::<Person, _>(&person_schema, &json)?;
```

For edges, the pattern is the same:

```
#[derive(Debug, Deserialize, Serialize, garde::Validate)]
#[garde(allow_unvalidated)]
struct WorksOn {
    #[garde(length(min = 1))]
    person_id: String,
    #[garde(length(min = 1))]
    project_id: String,
    #[garde(range(min = 1, max = 100))]
    allocation_percent: u8,
}

impl TypedEdge for WorksOn {
    const LABEL: &'static str = "WORKS_ON";

    fn source_node_id(&self) -> NodeId {
        format!("person:{}", self.person_id).into()
    }

    fn target_node_id(&self) -> NodeId {
        format!("project:{}", self.project_id).into()
    }
}

let works_on_schema = object()
    .field("person_id", string().min(1))
    .field("project_id", string().min(1))
    .field("allocation_percent", number().int().min(1.0).max(100.0))
    .strict();

builder.add_edge_from_json::<WorksOn, _>(
    &works_on_schema,
    &json!({
        "person_id": "nia",
        "project_id": "grust",
        "allocation_percent": 80
```

```
    }),
)?;
```

The adapter intentionally deserializes the original JSON after zod-rs accepts it. This preserves Rust integer types. For example, `zod-rs` validates numbers through a floating-point schema, but Serde should still be allowed to decode the original JSON integer `80` into a Rust `u8`.

5.4 How the Options Interact

There are four common construction paths:

```
Raw GraphBuilder:
  trusted Rust code -> Node/Edge -> Graph

Graph documents:
  YAML/JSON/XML -> Graph::from_* validation -> Graph

typed-garde:
  Rust struct -> garde validation -> TypedNode/TypedEdge lowering -> Graph

typed-zod-rs:
  raw JSON -> zod-rs shape validation -> Serde -> garde validation -> Graph
```

Choose `GraphBuilder` when the data is already trusted Rust code and the dynamic graph model is enough. Choose graph documents when you need readable fixtures, interchange files, or migration inputs. Choose `typed-garde` when the domain model lives in Rust structs and should enforce Rust-level invariants. Choose `typed-zod-rs` when the input is untrusted JSON and you want a separate shape gate before Serde and garde.

These options compose. A single graph can contain nodes loaded from YAML, raw edges added by `GraphBuilder`, typed nodes validated by garde, and request payloads validated first by zod-rs. The result is still just `Graph`.

The distinction between zod-rs and garde is worth keeping crisp:

- zod-rs sees `serde_json::Value`.
- Serde converts JSON into Rust types.
- garde sees Rust fields with Rust types.
- `TypedNode` and `TypedEdge` convert validated Rust values into Grust labels, ids, endpoints, and properties.

For example, this JSON fails at the zod-rs stage because `skills` is not an array:

```json
{
  "id": "nia",
  "name": "Nia",
  "skills": "rust"
}
```

This JSON can pass a loose zod-rs shape check but fail at the garde stage if the Rust type requires at least two skills:

```json
{
  "id": "nia",
  "name": "Nia",
  "skills": ["rust"]
}
```

That separation gives application authors precise error boundaries. Shape errors usually belong near the API or file-ingest boundary. Domain errors belong near the typed model.

5.5 Relationship to GraphSchema

`GraphSchema` and typed ingestion solve different problems. `GraphSchema` describes graph labels, fields, uniqueness, and backend-facing metadata. It can drive indexes, migrations, table layouts, or validation inside a store. Typed ingestion validates values before they become graph data.

In practice they can reinforce each other:

- `TypedNode` and `TypedEdge` keep application construction honest.
- zod-rs schemas protect JSON ingress points.
- `GraphSchema` tells storage backends what structure to expect.
- `GraphStore` remains the common persistence contract.

This layered design keeps Grust from becoming either too loose or too rigid. Projects can start with raw graph construction, add typed Rust validation for important labels, add zod-rs for JSON ingress, and later add `GraphSchema` for backend optimization.

The same schema can also validate and write in one step:

```rust
let report = store.put_typed_graph(&schema, &graph).await?;
```

That call validates labels, required fields, field value types, and edge endpoint labels before delegating to the backend. Schema-capable stores then use `apply_schema` to lower the portable model into their native storage surfaces.

6 5. Traversal as an Intermediate Representation

Grust traversal is not Cypher, SQL, GQL, SurrealQL, Spark SQL, or a graph database dialect. It is a small Rust intermediate representation:

```rust
let traversal = Traversal::from_node("talk:rust-graph-api")
    .out("PRESENTED_BY")
    .to("Person")
    .limit(10);
```

A traversal has a start expression, a sequence of steps, and an optional limit. The start can be a single node, all nodes with a label, or nodes with a property value. Each step has a direction, an optional edge label, and an optional target node label.

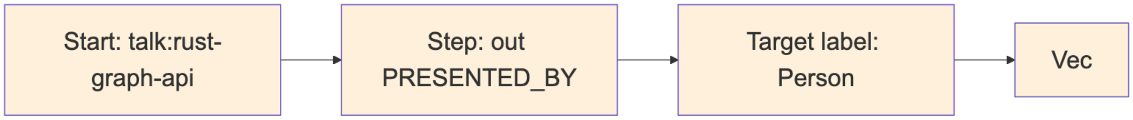

Figure 4: Diagram 4

The IR is deliberately modest. That makes it implementable across a wide range of backends. The memory backend can scan maps. LanceDB can query tables by hop. pgGraph can lower to SQL over universal graph tables. Sail can lower to Spark SQL through Spark Connect. Backends that do not yet support reads can return `GrustError::Unsupported`.

This design also protects application code. A caller asks for a graph-shaped operation; the backend decides whether that operation becomes a map scan, a SQL join, a DataFrame query, a Redis graph command, or a future native graph query.

For local analytics and backend planning, `grust-core` also exposes `GraphIndex`. It builds a dense vertex index from a `Graph`, validates that every edge endpoint exists, and stores incoming and outgoing edge indexes per vertex. That gives adapters and examples one shared adjacency layer instead of rebuilding node-id maps in each backend. The `grust-graph` facade includes a dependency-free `benchmarks` example that exercises graph cloning, `GraphIndex` construction, degree scans, endpoint scans, and structural edge-key generation over ring, grid, layered, clustered, Graph500-style R-MAT, and GAP-style R-MAT graph families.

7 6. The Store Contract

The central backend trait is `GraphStore` (capability and native-constraint methods are omitted here for brevity):

```
#[async_trait::async_trait]
pub trait GraphStore: Send + Sync {
    async fn apply_schema(&self, schema: &GraphSchema) -> Result<()>;
    async fn put_node(&self, node: &Node) -> Result<PutOutcome>;
    async fn put_edge(&self, edge: &Edge) -> Result<PutOutcome>;
    async fn put_graph(&self, graph: &Graph) -> Result<LoadReport>;
    async fn put_typed_graph(&self, schema: &GraphSchema, graph: &Graph) ->
Result<LoadReport>;
    async fn get_node(&self, id: &NodeId) -> Result<Option<Node>>;
    async fn get_nodes(&self, ids: &[NodeId]) -> Result<Vec<Node>>;
    async fn get_edges(&self, query: EdgeQuery) -> Result<Vec<Edge>>;
    async fn traverse(&self, traversal: Traversal) -> Result<Vec<Node>>;
}
```

The trait is async because real graph stores usually cross a process, network, database, or object-store boundary. The `async_trait` crate hides the current limitations around async functions in object-safe traits and lets backend implementations present one uniform interface.

`put_graph` takes `&Graph` instead of consuming `Graph`. That keeps retry, audit, comparison, and multi-backend loading workflows straightforward:

```rust
let report = store.put_graph(&graph).await?;
backup_store.put_graph(&graph).await?;
println!("loaded {} nodes and {} edges", report.nodes, report.edges);
```

Single-element writes return `PutOutcome`. Memory and builder paths can report precise inserted/updated/deduped outcomes. Remote upsert-oriented backends commonly return `Upserted` because distinguishing insert from update would require an extra read or a backend-specific write primitive. Portable callers should treat all written outcomes as success rather than depending on inserted-versus-updated. `LoadReport` counts elements that were written or upserted, not necessarily newly created rows.

Administrative operations are split into `GraphAdminStore`:

```rust
#[async_trait::async_trait]
pub trait GraphAdminStore: GraphStore {
    async fn bootstrap(&self) -> Result<()> { Ok(()) }
    async fn clear(&self) -> Result<()>;
}
```

This separation keeps ordinary application persistence apart from setup and destructive workflows. A production service may receive a `GraphStore`, while a test harness or migration tool can require `GraphAdminStore`.

8 7. Rust Concepts Used

Grust is a good example of idiomatic Rust applied to a storage abstraction.

Newtypes give semantic weight to strings. `NodeId`, `EdgeId`, and `Label` are cheap wrappers, but they prevent accidental parameter swaps and produce clearer APIs.

Trait-based polymorphism defines the backend boundary. Application code can be generic over `impl GraphStore`, while each backend owns its own connection, configuration, batching, serialization, and query strategy.

Feature flags keep the facade crate light. The public `grust` crate re-exports backend crates only when features such as `memory`, `lancedb`, `postgres`, `postgres-pgq`, `pggraph`, `turso`, `sail`, `falkor`, `surreal`, or `cocoindex` are enabled. The HelixDB and LadybugDB adapters remain internal `publish = false` workspace crates and are deliberately absent from the crates.io facade.

Serde makes the graph model portable. Core types derive `Serialize` and `Deserialize`, and backends can turn properties into JSON strings, JSONB, Arrow string columns, or target-state maps.

Interior mutability appears where it is appropriate. The memory backend stores its graph behind `Arc<RwLock<...>>`, making cloned stores share state while keeping mutation synchronized.

Error typing is explicit. `GrustError` distinguishes backend, schema, unsupported-feature, and serialization failures. That lets a caller tell the difference between “the database rejected this” and “this backend does not yet support traversal.”

9 8. Backend Architecture

Backends share the same input model but not the same execution model:

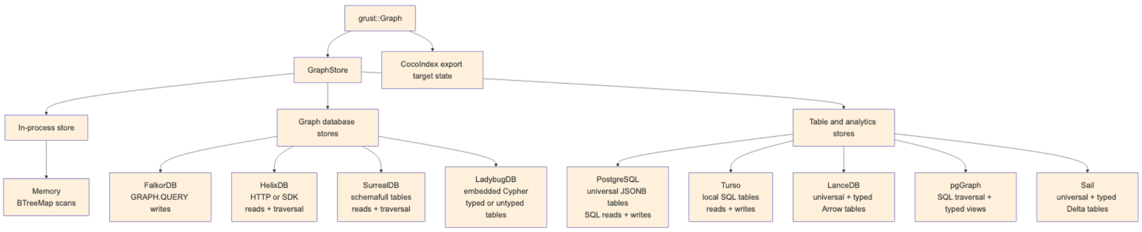

Figure 5: Diagram 5

Some backends are full read/write/traversal stores today. Others still focus on writes and administrative loading. That is normal for an early multi-backend project, and the trait makes the maturity boundary explicit.

9.1 Memory

`grust-memory` is the deterministic local backend. It stores nodes in a `BTreeMap<NodeId, Node>` and stores edges by source, label, destination, and optional explicit edge ID. That means structural edges still behave deterministically, while id-bearing parallel edges between the same endpoints can coexist. Reads and traversals scan those maps. It is the best backend for tests, examples, and local workflows that need no external service. When a `GraphSchema` is applied, the memory backend validates writes against it, which makes it a useful conformance harness for typed storage behavior before a database enters the picture.

```
use grust::prelude::*;

# async fn demo(graph: Graph) -> grust::Result<()> {
let store = MemoryGraphStore::new();
store.put_graph(&graph).await?;

let people = store
    .traverse(
        Traversal::from_node("talk:rust-graph-api")
            .out("PRESENTED_BY")
            .to("Person"),
    )
    .await?;
# Ok(())
# }
```

9.2 LanceDB

`grust-lancedb` treats LanceDB first as a durable Arrow-native table store. It uses two tables, one for nodes and one for edges. Nodes are keyed by `id`. Edges are keyed by an explicit edge id when present, or by a deterministic `from + label + to` key otherwise. That fallback key is shared with other tabular/export backends through `grust-core::edge_key`, so structural edge identity does not drift across implementations. Writes use LanceDB `merge_insert`, and traversal performs hop-by-hop table queries. When a hop fans out to multiple target nodes, the backend reads those target nodes with `get_nodes` instead of issuing one node query per edge. Property-start traversal filters by label in LanceDB, then compares decoded Grust properties exactly so nested JSON or serialized fragments cannot produce false positives. Before persisting either an explicit or structural edge key, LanceDB calls the checked identity path and rejects U+001F in the source ID, label, target ID, or explicit ID. CocoIndex, LadybugDB, Sail, and Cypher capture/refetch use the same guard, and mixed explicit/idless comparisons also require the same structural owner instead of trusting a key-shaped string alone. Ladybug's managed metadata index reserves the same delimiter, so its adapter rejects U+001F in node IDs before a user record can alias a table marker.

Schema object identity is checked separately. The shared `validate_physical_identifier_claims` helper lets FalkorDB, Helix, LadybugDB, LanceDB, and Sail reject lossy-name collisions and exact duplicate declarations within each native namespace before schema or write operations are emitted.

This backend is a natural home for later vector-search extensions. The core `GraphStore` trait should stay graph-focused; LanceDB-specific nearest-neighbor search can live in an extension trait without leaking into every backend.

With `GraphSchema`, LanceDB also creates typed Arrow tables per node and edge label. The universal `grust_nodes` and `grust_edges` tables remain the portable read/traversal surface, while schema-labeled rows are mirrored into tables such as `grust_node_person` or `grust_edge_presents` with typed columns for declared fields. That gives analytical consumers and future vector extensions a native columnar surface without giving up the backend-neutral graph model.

9.3 LadybugDB

`grust-ladybug` embeds LadybugDB directly through the Rust `lbug` 0.20.2 crate. It is the durable local graph-database backend: no Docker service, no HTTP bridge, and no separate daemon. The store opens either an in-memory Ladybug database or an on-disk Ladybug directory and creates Grust-managed Ladybug node and relationship tables from graph labels.

Ladybug can be used for typed or untyped property graphs, and the Grust backend now exposes that distinction directly. The default `LadybugGraphMode::Untyped` accepts ordinary Grust graphs and creates the needed Ladybug node and relationship tables from graph labels and endpoint labels on write. That mode matches Grust's raw `Graph`, JSON, YAML, and XML loading path, where labels and properties are data owned by the application.

`LadybugGraphMode::Typed` requires `apply_schema` or `put_typed_graph` before writes. In that mode the backend creates the declared Ladybug tables up front, does not create undeclared tables during writes, and validates later writes against the applied `GraphSchema`. Reads reconstruct Grust `Node` and `Edge` values from the managed tables, while traversal evaluates the portable Grust traversal IR by walking Ladybug relationship tables and reading target nodes through the same `GraphStore` contract.

Ladybug's managed metadata index uses U+001F to frame table markers and node entries. The adapter therefore rejects that delimiter in node IDs before any mutation; this is stricter than the backend-neutral `NodeId` type and is part of the Ladybug storage contract.

The internal crate's `arrow` feature also exposes Ladybug's embedded Arrow table path through Arrow IPC streams. A workspace caller can register IPC node tables, relationship tables, and CSR relationship tables directly with Ladybug, then query them with Ladybug Cypher and receive result chunks back as Arrow IPC. The public boundary is IPC bytes rather than a Rust `RecordBatch` type, so callers do not have to match Ladybug's internal Arrow crate version exactly.

The first implementation stores Grust properties as JSON text for portable round trips. Later schema lowering can add typed Ladybug columns, full-text indexes, vector indexes, and direct graph-RAG extension traits without changing the core graph model.

9.4 PostgreSQL and pgGraph

`grust-postgres` stores the source of truth in ordinary PostgreSQL tables:

```
grust_nodes(id text primary key, label text not null, props jsonb not null)
grust_edges(id text, from_id text not null, to_id text not null,
            label text not null, props jsonb not null)
```

The generic backend creates tables and indexes with ordinary PostgreSQL DDL. It does not require PostgreSQL extensions, so the same `PostgresGraphStore` can target local PostgreSQL, Neon, or another managed PostgreSQL-compatible service. Reads use SQL against the universal tables. Traversal is lowered to SQL joins over those tables. Mutation batches are wrapped in PostgreSQL transactions.

The shared connection is serialized across every explicit transaction. A recovery marker is set before `BEGIN` and cleared only after PostgreSQL acknowledges `COMMIT` or `ROLLBACK`; if cancellation drops a future mid-flight, the next serialized caller rolls back the uncertain transaction before doing new work. The public raw `PostgresGraphStore::execute` surface is deliberately autocommit-only. Its lexical guard rejects transaction-control statements in a batch while ignoring lookalike words inside strings, identifiers, dollar-quoted bodies, and comments. PostgreSQL PGQ forwards through the same contract.

Schema application adds typed label views and expression indexes. For example, a `Person` node schema with `name: String` and `age: Int` can produce a `grust_node_person` view over the universal node table, with `age` exposed as a `bigint` expression. This is a deliberately incremental typed-storage path: PostgreSQL keeps the flexible JSONB source of truth while callers that know the schema get typed SQL surfaces.

`grust-pggraph` is now the pgGraph extension layer over that same shared PostgreSQL implementation. It bootstraps the `graph` extension, registers the universal tables, and can optionally build the pgGraph projection for graph-index experiments without forking the storage and traversal code.

`grust-postgres-pgq` targets PostgreSQL 19's native SQL/PGQ support. It uses the same universal tables as the durable storage layout, creates a native `PROPERTY GRAPH` over them, and executes bounded traversal through GRAPH_TABLE. That keeps writes, reads, schema views, and mutation batches on the proven PostgreSQL backend while letting traversal exercise PostgreSQL's standard property-graph query engine.

The reusable SQL boundary lives in grust-sql-core. It owns the parts that are really common across row-store SQL graph backends: universal table DDL, reads, bounded traversal joins, mutation framing, schema views, expression indexes, identifier quoting, and literal escaping. Dialects keep the pieces that affect correctness or efficiency. PostgreSQL keeps JSONB predicates, ON CONFLICT, CREATE OR REPLACE VIEW, transaction text, and lateral joins for undirected steps. Turso keeps JSON text, json_extract, json_patch, SQLite-compatible views, and a derived-table undirected join shape. Sail does not use this SQL core because its SQL path runs through Spark Connect, Arrow IPC staging, and distributed Spark SQL rather than a direct row-store connection.

Recursive walk pushdown no longer stores raw node IDs between sentinel delimiters. PostgreSQL, Spark SQL, and the generic SQLite dialect encode IDs as hexadecimal, delimiter-free tokens before constructing visited sets. A dialect without both recursive-CTE support and an encoding hook declines variable-path or shortest-walk pushdown and preserves correctness through fallback.

GraphSqlDialect::max_identifier_bytes lets a dialect declare a limit for generated schema identifiers. PostgreSQL reports its 63-byte ceiling: typed node/edge view and property-index names at the limit remain valid, while longer names fail with GrustError::Schema before the server can silently truncate them into an ambiguous or colliding identifier. Dialects that retain the default None keep their existing behavior.

9.5 Turso

grust-turso uses the Turso Rust SDK directly. The default connection path opens a local in-process Turso database:

```
# use grust::prelude::*;
# async fn open() -> grust::Result<()> {
let store = TursoGraphStore::connect(TursoConfig {
    path: "data/grust.db".to_string(),
    table_prefix: "grust".to_string(),
    batch_size: 500,
    journal_mode: TursoJournalMode::Wal,
})
.await?;
# Ok(())
# }
```

The storage layout mirrors the universal table shape used by the PostgreSQL backend, but uses SQLite-compatible types and JSON text properties:

```
grust_nodes(id text primary key, label text not null, props text not null)
grust_edges(id text, from_id text not null, to_id text not null,
            label text not null, props text not null)
```

Reads and traversal use ordinary SQL over the local Turso connection. Schema application creates label-specific SQL views and expression indexes using `json_extract`. Mutation batches are wrapped in a Turso transaction.

The `journal_mode` option selects the concurrency model. The default `Wal` is Turso's single-writer write-ahead log. Selecting `Mvcc` enables Turso's multi-version concurrency control (`PRAGMA journal_mode = mvcc`, a database-header mode applied to a fresh database); data writes then run inside `BEGIN CONCURRENT` transactions with bounded conflict retry, so concurrent writers make progress.

With the `turso-sync` facade feature, callers can construct a synced store from a local path, remote URL, and optional auth token. The `GraphStore` API continues to operate on the local SQL connection; callers decide when to call the store's `push` and `pull` helpers.

9.6 QueryGraph Memory: TypeSec over Grust

`querygraph-memory` is a concrete application of the Grust backend contract. It implements TypeSec's synchronous `MemoryStore` interface over a compatible Grust mutation backend without moving authority or plaintext handling into the storage layer:

```rust
pub struct GraphStoreMemoryStore<G: GraphMutationStore> { /* ... */ }
```

```rust
use querygraph_memory::TursoMemoryStore;

let store = TursoMemoryStore::open("data/querygraph-memory.db")?;
```

`GraphStoreMemoryStore<G>` is the generic adapter for an already-initialized `GraphMutationStore`. The production-oriented v1 alias, `TursoMemoryStore`, binds that adapter to `TursoGraphStore`. `TursoMemoryStore::open(path)` creates missing parent directories, opens a file-backed local database with the stable `querygraph_memory` table prefix, and runs Grust's bootstrap before returning. `open_with_config` is the explicit form for applications that need another table prefix, batch size, path, or journal mode. A default `TursoConfig` still uses `:memory:`, so durable callers must supply a file path.

The adapter projects memory into a small property graph:

```
(:MemoryRecord {record: <opaque JSON>, space: ...})
    -[:MENTIONS]->(:MemoryEntity {name, kind})
(:MemoryEntity)-[:RELATES]->(:MemoryRelation {rel, fact_id})
    -[:RELATES]->(:MemoryEntity)
```

Each `MemoryRelation` is an assertion, not merely an endpoint pair. Its node ID is a SHA-256 identity over length-prefixed source, relationship name, target, and record ID components. Replaying one record is stable, while two records that assert the same named relation between the same entities retain separate lineage. Neighborhood traversal follows the two-edge assertion shape and also reads the legacy direct `RELATES {rel, fact_id}` representation so existing durable stores remain compatible. Tombstone preflight first discovers the record's assertion nodes and any legacy fact edges, then submits deletion of that discovered set with the record node as one mutation batch. A transactional backend makes that deletion batch atomic, and other records' assertions survive. The discovery reads are not isolated inside that transaction: callers must synchronize a tombstone with concurrent links for the same record so a new assertion cannot appear after discovery and escape the deletion batch.

The complete TypeSec `StoredRecord` is serialized into one opaque JSON property and round-tripped whole. Grust does not open the protected content. TypeSec's `MemoryVault` remains the only component that rehydrates it, verifies typed capabilities, applies a recall clearance ceiling, excludes quarantined records, joins sensitivity labels during consolidation, and emits audit evidence. That separation is the security boundary: Grust supplies durable graph mechanics; TypeSec decides which subject may learn what.

Queries push only memory-space equality into the graph as `Start::NodesByProperty`. The other `StoreQuery` dimensions—kind, time, label, entities, text, invalidation state, and ordering—continue through TypeSec's shared `StoreQuery::matches` implementation. This is semantically complete and conformance-tested, but it is deliberately not advertised as full GQL pushdown. A neighborhood traversal may discover record identifiers associated with global entity nodes across several spaces; the vault rejects every record outside the authorized space before content can be revealed. Tenant isolation in v1 is therefore authorization at the vault boundary, not physical graph partitioning.

Consolidation uses the mutation boundary rather than a special memory-only transaction API. `MemoryStore::apply_batch` converts its puts and invalidations into one `GraphMutation` slice and calls `apply_mutations` once. Turso reports transactional mutation atomicity, so superseding old records and inserting the replacement commits as one unit. TypeSec performs the SecLib label join before that batch reaches storage, which means a replacement derived from a Sensitive source remains Sensitive after closing and reopening the database. A generic backend that reports `OrderedNonAtomic` remains usable for simpler operations but cannot promise atomic consolidation.

The adapter also owns the sanctioned bridge between TypeSec's synchronous `MemoryStore` and Grust's asynchronous `GraphStore`. A dedicated current-thread Tokio runtime carries I/O and time drivers. Calls made outside Tokio drive it directly; calls made from an existing Tokio runtime execute on a scoped thread. This keeps an MCP or HTTP service from nesting runtimes or panicking, and lets the store shut down safely when dropped from asynchronous application code.

Semantic ranking and cognition preserve the same boundary. The reference `VectorIndex<E: Embedder>` is an in-process cosine index. An `Embedder` declares whether it is local; a remote embedder is never handed Sensitive or Secret text, and those records remain available to ordinary authorized recall without participating in remote vector ranking. Search returns candidate IDs, with an optional bounded entity co-mention boost, but the vault still performs the authorization and label checks. Reference deduplication, contradiction, and importance analyzers likewise make no writes. They consume already-recalled views and emit inert `ConsolidationPlan` values that must return through the capability-gated vault.

The governed cognition path extends that rule instead of bypassing it. A `CognitionRequest` contains one TypeSec-authorized input, its canonical binding, its optional vault-verified governed source scope, the exact LakeCat snapshot and policy-narrowed projection, the field mapping used by governed ingestion to derive the selected memory records, a durable job identity, and either the deduplicate or reconcile operation. The trusted host selects a fixed reference or native Sail profile before protected input is loaded; public engine implementations cannot report their own trusted identity. Both asynchronous engines return a bound `CognitionProposal`, never a store handle or direct write.

Every bound proposal uses TypeSec proposal schema version 4. That wire contract identifies `input_snapshot` with the canonical immutable snapshot digest, keeps the LakeCat grant digest as separate binding evidence, and carries an explicit `mutated` or `no_change` effect. Reference and Sail derive that effect from the complete proposal: zero drafts and zero plan steps means no-change; every nonempty plan is mutating. Earlier bound schema versions fail closed; unbound schema version 1 remains only an inert local planning value. This wire version is independent of the operation algorithm version below.

Deduplicate and reconcile each own an explicit version-2 semantic contract. Reference and Sail profiles bind the same per-operation version because they must produce the same canonical plan. Crate, package, and build versions remain useful implementation metadata, but never substitute for the algorithm version in signed TypeDID authority. Previously signed version-1 or package-bound intents are deliberately incompatible and must be authorized again; no native profile silently upgrades their authority.

Live Sail does not re-read LakeCat rows or reinterpret the ingestion mapping. It derives and stages only authorized IDs, normalized text keys, contradiction prefix/tail keys, and validity timestamps under a collision-safe session view; it does not stage the raw text column. Planning has independent finite operation, abort, and cleanup deadlines, and cleanup is attempted after success, failure, timeout, or caller cancellation. The Spark client and cognition decoder reuse the same public 16 MiB Arrow IPC payload limit; the decoded protobuf message has one additional MiB of bounded envelope headroom. Normalized and contradiction keys remain content-derived rather than anonymized, so the Sail endpoint belongs inside the processing boundary authorized for that protected input. Arrow framing, schemas, declared rows, buffers, compressed expansion, result counts, and local reconciliation work are checked before Arrow result-array allocation and then rechecked against the complete result. The reference and Sail engines use the same deterministic planning functions, so input permutation and timestamp ties produce canonical output.

Durability is a separate storage capability. `GraphCommitStore` combines exact node or absence expectations, a mutation batch, an idempotency digest, and a backend receipt in one transaction. Turso mints the receipt's canonical UTC RFC 3339 timestamp at nanosecond precision immediately before inserting that receipt into the same transaction. It is backend-issued transaction-boundary evidence, not a wall-clock observation taken after storage fsync, and recovery returns those exact persisted bytes or fails closed on malformed time. The cognition scheduler stores only scoped digests, issues bounded renewable bearer leases, persists only the canonical proposal digest, and survives reopen. During application, TypeSec supplies the opaque prepared commit; Grust atomically checks source revisions, applies its exact memory operations and ID-only index outbox, writes the audit record, persists the outcome, and completes the job. A no-change token has empty operations, affected IDs, and outbox, and retains the prior memory version, but the exact source guards, job, audit, outcome, and guarded ledger still commit as one decision. Recovery is read-only and cross-validates the job, audit, outcome, authority scope, optional governed source scope, proposal, effect, and backend receipt. Durable outcome schema version 3 requires TypeSec audit schema version 2, which carries the same typed effect, distinct grant and snapshot digests, and the trusted authority-revalidation and preparation times. Audit and commit-envelope digests use version-3 domains so this evidence layout cannot be confused with either predecessor. TypeSec owns scope selection and authoritative reload checks; Grust preserves that evidence and atomically enforces the prepared full-record preconditions. Tests exercise concurrent identical decisions and commit-then-response loss and prove that retry plus reopen retain exactly one job, audit record, outcome, guarded ledger, and the exact mutation/outbox shape required by the effect.

The scheduler's `transitionedAt` is a caller-supplied logical transition time; for `Completed` it is explicitly the TypeSec audit's `preparedAt`. It is not a backend commit timestamp. A completed job's `completionDigest` is exactly the canonical TypeSec prepared digest for either effect, never the resulting memory version; this avoids collapsing no-change into an unchanged memory version. Recovery checks durable schema versions before deserialization, rejects incompatible historical outcome and audit layouts, and requires affected IDs to retain TypeSec's strict canonical order. Authoritative `committedAt` evidence exists only in the outcome and receipt, must be canonical RFC 3339, and cannot predate preparation. Grust rejects malformed or regressive backend time on initial return and recovery instead of substituting a timestamp from another phase.

The scheduler and outbox APIs are storage primitives behind Marciana's authenticated scheduler and trusted worker pool. A worker intentionally need not equal the submitter, which lets expired work move safely, but canonical owner strings and scoped job keys are not credentials. Once issued, lease and claim tokens are bearer credentials for worker transitions. Marciana must authorize acquisition and cancellation and keep those tokens confidential; only a freshly prepared opaque TypeSec commit can authorize memory mutation.

This is a native Grust, TypeSec, LakeCat, and Sail composition. Cognee supplied design inspiration only; no Cognee runtime, adapter, or storage dependency is present.

The durable proof still has explicit limits. `MemoryId::next()` uses a process-local counter, so a restarted ordinary writer can collide with persisted `mem-N` identifiers; a hosted or multi-process service should mint collision-resistant IDs at its boundary. `VectorIndex` is not a persistent LanceDB ANN implementation, and memory predicates beyond space are not fully pushed into GQL. Cognition job idempotency does not replace durable TypeDID nonce replay protection shared across gateway replicas. Finally, vault-level tenant checks, restart persistence, and running-service conformance do not by themselves constitute a hosted multi-tenant service with quotas, migrations, backups, deletion propagation, and service-level objectives.

9.7 Sail

`grust-sail` connects to a Sail Spark Connect server over gRPC. It stores graph data in Spark DataFrames backed by Delta tables. SQL commands and reads are sent as Spark Connect SQL relation plans, and read results are decoded from Arrow IPC streams.

Portable count projections may lower to `SELECT 1` when no named binding must be returned. Sail sends that row-presence marker as an Arrow integer rather than a string. Grust decodes integer markers alongside text columns, preserving one row per match for the shared Rust projection; it does not mistake this path for a server-side aggregate. This boundary is regression-tested for multiple result batches, nulls, and empty results. Sail 0.7.1 cannot execute the generated recursive walk CTE, so variable-length path reads use the shared reference fallback. Benchmark evidence labels this as graph materialization plus Rust execution, not native Sail path performance; downloaded-scale admission may reject that materialization path.

Call `SailGraphStore::close().await` after a session's operations finish to release its remote temporary views and session state. This consumes the Rust client and invalidates any other client sharing that session ID; durable warehouse files are not deleted. Ordinary Rust drop cannot perform this asynchronous cleanup. Bound the close future if your application requires a deadline, and treat timeout or failure as uncertain cleanup. A release acknowledgement alone does not prove an interrupted query has stopped.

Connection establishment validates the client configuration. The default `SailWarehouse::ServerManaged` policy does not set `spark.sql.warehouse.dir`; Sail's catalog and warehouse configuration remain authoritative. Connection failures use a stable message and do not render the configured endpoint or transport error because either can disclose endpoint credentials or signed parameters. The warehouse policy is safe across a remote client boundary and does not silently select a new client-local persistence path. Co-located development can opt into `SailWarehouse::LocalSessionScoped`, which derives a path beneath the client's temporary directory from the session ID. Grust does not delete that directory; callers own cleanup, and reusing the session ID reuses the path. Durable callers can use `SailWarehouse::ExplicitPath` with a stable absolute path visible to the server; Grust sets and reads that override back through the same Spark Connect session. Reopening tables across sessions additionally requires Sail to provide persistent catalog metadata. If Sail's unconfigured warehouse fallback is the relative `spark-warehouse` path, the server needs an absolute setting or the client must select one of the explicit Grust overrides before creating managed Delta tables.

Bulk writes stage Arrow IPC batches as Spark Connect `LocalRelation` temp views and then merge from those views. That avoids building one giant SQL literal per row, keeps user values out of SQL text, and gives long-running requests an operation id with reattachment enabled. Query filters bind user values through Spark Connect named arguments. Delete mutations stage their values as Arrow temp views before running argument-free SQL commands, which avoids string substitution while matching Sail's current command-parameter behavior. Traversal joins use globally unique node ids; source and destination label columns may be empty for single-edge writes where the full graph is not in scope.

Sail's Arrow boundary is now public too. Applications can stage arbitrary Arrow IPC streams as session temp views and query them with Spark SQL, collect Spark SQL results as Arrow IPC chunks, or load Grust-shaped node and edge IPC streams through the normal graph write path. This gives Sail the same data-source role as Ladybug while keeping Sail's internal Arrow 58 dependency separate from Ladybug's Arrow 55 dependency. Staged views can be dropped through a validated, idempotent helper, allowing protected batch inputs to be cleaned up on success, execution failure, or an uncertain retry.

When a schema is applied, Sail creates typed Delta tables per node and edge label and mirrors writes into them with `MERGE INTO`. The universal Spark tables keep traversal simple and portable; the typed tables make declared graph labels available as ordinary Spark columns. Their declared names survive table creation, while Delta constraints reject null structural node and edge identities. Constraint-registry values likewise enter through staged Arrow rather than SQL string interpolation.

Sail also has reusable graph analytics helpers over the persisted generic tables. `read_graph` collects the generic `grust_nodes` and `grust_edges` tables back into a portable Grust `Graph`. `in_degrees`, `out_degrees`, `degrees`, and `degree_pairs` run Spark SQL over those same tables and decode the Arrow results into small Rust row types. These helpers are deliberately low-level: they expose common graph measurements without making the backend-neutral `GraphStore` trait depend on Spark-specific analytics.

The Sail crate now also publishes its generic table contract. Constants name `grust_nodes`, `grust_edges`, and the physical node and edge columns, including the persisted generic edge `edge_key` and optional explicit edge `id`. Projection helpers classify requested graph fields as physical columns or JSON properties, map edge `label` to the stored `edge_type`, and check when a typed Sail node or edge table can satisfy a graph query directly. This keeps Sail-native graph planning and GrustFrames-style distributed lowerings aligned with the same backend layout that Grust writes.

The same contract layer exposes typed table descriptors derived from `GraphSchema` and directional triplet SQL. The triplet helpers join generic edges back to their source and destination nodes, and can orient rows as outgoing, incoming, or undirected pairs. That is the shared primitive needed by distributed triplet filters, motif expansion, and aggregate-message style passes without making the backend-neutral store trait depend on those higher level algorithms.

Writable Cypher is not a separate Sail persistence path. The portable Cypher/GQL layer (see *Cypher and GQL*) owns parsing, semantics, planning, and the read engine; for the bounded read subset it lowers the `MATCH`/`WHERE` filter into Spark SQL while the `RETURN` projection runs through the shared reference. `SailGraphStore` executes resolved `GraphMutationPlan`s through `CypherMutationExecutor`, so Cypher writes use the same staged-Arrow, `MERGE INTO`, typed-Delta mirror, and delete paths as ordinary Grust writes. The plan and report types are backend-neutral — Memory, Sail, and Turso all execute them — while Sail adds the SQL read pushdown and typed-table mirrors on top.

9.8 FalkorDB, HelixDB, and SurrealDB

The FalkorDB backend writes through Redis `GRAPH.QUERY` using Cypher-like `MERGE` statements. It batches nodes by label path and edges by relationship type. Schema application creates label/property indexes for declared node types. Configurable identity-property names and generated labels, relationships, and properties are validated before Cypher construction. Property names retain their physical spelling through backtick quoting where FalkorDB permits it; unsafe delimiters and normalized-name collisions within a schema or complete graph load fail closed. The configured structural ID wins over property-map data, and connection-pool/query failures do not render the Redis URL or credentials.

The HelixDB backend has HTTP and SDK stores. Both support batched writes, node reads, edge reads, and backend-neutral traversal through Helix dynamic queries. Edge writes store Grust relationship metadata (`relationship`, `from_id`, `to_id`, and optional `edge_id`) so `EdgeQuery` can reconstruct Grust edges from Helix relationship rows. Helix writes preserve supported scalar and array properties instead of silently dropping non-string values; unsupported JSON object properties return an explicit error. The current schema hook validates that labels, relationships, and fields can be safely lowered through the dynamic-query path; backend-native schema-file generation can build on that same `GraphSchema` contract later. Schema and graph preflight also reject normalized relationship collisions and attempts to declare or write the structural `id`/label and edge-metadata fields. Both HTTP and SDK graph loads validate all chunks before transport, and transport failures omit configured URLs and their embedded credentials or query strings.

The SurrealDB backend also has HTTP and SDK stores. It can bootstrap, clear, upsert nodes, relate edges, delete nodes and edges, read nodes and edges, and execute traversal hop-by-hop through the `GraphStore` contract. Applying a schema lowers node and edge declarations to Surreal `DEFINE TABLE` and `DEFINE FIELD` statements so the backend can run in schemafull mode where the schema calls for it. Reads use configured labels and relationships, plus ID-derived table names, to find records across Surreal's label-specific tables. Generic edge reads and node deletes require `SurrealConfig.relationships`; when that list is empty, the backend returns a configuration error instead of silently scanning no relation tables. Explicit edge-label reads and deletes can still target a known relation table directly. Traversal batches target-node reads per step through `get_nodes`, avoiding a serial node lookup for every edge in a fan-out. Mutation batches are wrapped in SurrealDB transactions. SurrealDB 3.2 identifiers are quoted without lossy property-name rewriting. Every newly written node stores its original, case-sensitive Grust label in the reserved `__grust_label` field instead of reconstructing that label from a normalized, lowercase physical table. Reads of older rows fall back to the physical table label exposed as `__grust_physical_label`. Record decoding separates the table at the first colon and removes only a matching outer backtick pair, preserving colon-bearing logical IDs such as `City:4` without a trailing backtick. Configuration, schema fields, normalized node/relation table claims, reserved storage fields, and complete graph batches validate before bootstrap or write I/O. Optional Grust edge IDs are stored separately as `edge_id`; node `id` and `labels`, relation `in`/`out`, and internal metadata cannot be overwritten by user properties. HTTP and WebSocket failures omit URL userinfo and query secrets.

9.9 CocoIndex

`grust-cocoindex` is intentionally not an ordinary `GraphStore`. CocoIndex is an incremental target-state system, so the Grust integration exports a graph as serializable node and relationship state:

```
let export = graph.to_cocoindex_export()?;
let graph = cocoindex_export_to_graph(export)?;
```

The export uses Grust node IDs as target keys, converts properties to JSON, and requires edge endpoints to exist in the graph so relationship source and target labels can be emitted. The import path expects the same key shape, validates relationship endpoints against imported nodes, and preserves explicit relationship keys as Grust edge IDs.

9.10 Backend Integration Tests

Unit tests stay self-contained, but live backend tests are explicit. They are marked ignored in Cargo so a normal workspace test run does not accidentally depend on local services. When a live test is requested, however, it must reach the backend; it no longer returns early and pretends success when a server is missing.

The repository provides a launcher for those checks:

```
scripts/integration-test.sh doctor --profile docker --mode docker
scripts/integration-test.sh --profile docker --mode docker
```

The Docker profile is the contributor path: it starts the Docker-backed services and runs local LanceDB and CocoIndex checks. The full maintainer profile is still available:

```
scripts/integration-test.sh --profile all
```

The launcher reads `integration/backends.conf`. In `auto` mode it prefers already-running services, then configured local source checkouts such as `~/src/sail`, `~/src/SurrealDB`, `~/src/FalkorDB`, and `~/src/HelixDB`, then Docker Compose where a service is available. The repository-level `docs/INTEGRATION.md` guide covers profiles, modes, Docker image pins, source-checkout configuration, and CI strategy.

Krill 0.13.2 is a scoped registry patch: `grust-cypher`, `grust-sail`, and the `grust-graph` facade move to 0.13.2. Surreal remains 0.13.1, while consumers naming any other published Grust crate directly continue to use 0.13.0. Krill corrects Sail integer row-marker decoding and recursive-path admission, and adds explicit remote-session cleanup. It retains Crayfish's endpoint-safe connection errors and faithful Surreal logical identities, and inherits Prawn's dependency qualification: Redis client 1.6.0 and FalkorDB service v4.20.4, SurrealDB Rust SDK and service 3.2.4 with reqwest 0.13.4, pgGraph service 1.2.0, tokio-postgres 0.7.18, and stable Turso 0.7.2. These are tested compatibility updates, not claims that every adapter implements the portable Cypher executor.

LanceDB stays at 0.30.0: the attempted 0.38.0 default-feature local build fails within upstream `lancedb` because `job.rs` references the remote-only `Error::Http` variant when `remote` is disabled. The unpublished Helix adapter now targets exact `helix-db` 3.0.0: its SDK path uses typed nested-AST `QueryRequest` builders and `Client::query(request)` rather than `DynamicQueryRequest` and `dynamic_query`. The direct HTTP/v1 store remains separate and unchanged. Nineteen unit tests pass, including SDK serialization and existing HTTP behavior; live SDK/v2 service qualification remains pending. Historical HTTP service evidence is not reused as proof of SDK/v2 compatibility. The repository's `benchmarks/lsqb/BACKENDS.md` records the full qualification matrix and live gate evidence.

9.11 LSQB Compatibility Microbenchmark

The repository also carries a Docker-reproducible compatibility workload in `benchmarks/lsqb`. The unmodified upstream side pins Graph Data Council LSQB commit `242cb2fd31340ca688954cb94794d74c0d5b6f92`, LadybugDB 0.19.0, and a digest-pinned Python 3.12.11 container. It validates LSQB's nine count queries independently of Grust.

The adapted Grust side is rectangular across twelve declared backends: Memory, Turso, PostgreSQL, Ladybug, FalkorDB, SurrealDB, LanceDB, Sail, pgGraph, PostgreSQL PGQ, Helix, and CocoIndex. Baseline and adversarial suites each contain one cell per backend, yielding 24 count reports in a complete run even when a declared cell is unsupported, unavailable, or not applicable. For every backend, the baseline cell declares nine LSQB-derived queries and the adversarial cell declares 13 separately labeled count attacks. A distinct backend-neutral suite has 14 required policy rejections, so the adversari.al extension contains 27 attacks across two non-overlapping expectation models.

The executable additionally exposes distinct `helix-sdk` and `surreal-sdk` network-client lanes, while the historical twelve-backend launcher and receipts remain unchanged. Both Helix and Surreal retain their direct HTTP lanes. Surreal's separately published Rust SDK/WebSocket example cohort passes 108 baseline and 156 adversarial observations, with two warm-ups and ten measured repetitions per query in rotating order. It measures backend materialization plus Rust reference execution, not native Surreal query-engine performance. Load, worker setup, query, and recovery timings remain separate. Helix's SDK3 source-built `/v2/query` service is still undergoing live qualification; its public build guide and candidate recipe do not establish a passing result.

Reports name the measured execution class as `in-process-reference`, `backend-native-aggregate`, `backend-row-source-rust-projection`, `backend-materialize-rust-reference`, or `backend-neutral-policy`. Each query ends as `pass`, `mismatch`, `unsupported`, `unavailable`, `timeout`, `error`, or `not_applicable`; policy cases separately record pass/fail and a stable rejection category. A capability or service gap remains visible rather than becoming a fallback pass. Those execution classes are not performance equivalent and their timings must not be collapsed.

The 28-node, 72-edge `sfexample` graph is a conformance and orchestration gate, not a backend ranking. Authenticated LSQB SF0.1 and SF0.3 downloads add larger tiers with sealed archive and extracted-manifest identities. At those scales, the harness admits in-process reference, backend row-source with disclosed Rust projection, and backend-native aggregate execution, while whole-store materialization plus the Rust reference is explicitly `unsupported`. The manifest additionally binds a per-query, per-execution-class logical-row bound: only exact cardinalities or certified upper bounds at or below 1,000,000 are timed on downloaded tiers. Larger or insufficient Rust-row bounds are explicit unsupported outcomes without samples, while native scalar aggregates remain a separate class. The fourteen-case policy suite remains fixed to `sfexample`. Sail, PostgreSQL PGQ, and Helix have no default pinned service startup contract; their cells are `unavailable` unless an operator explicitly qualifies a digest-pinned, resource-limited external service.

This is a conformance and reproducibility microbenchmark, and LSQB is not an official LDBC benchmark. A complete Grust matrix requires a valid publication receipt; separate native Neo4j, source-built Sail and Surreal SDK cohorts use immutable evidence bundles admitted by an independent site verifier. A bundle manifest's hashes alone do not establish qualification. Unadmitted diagnostic and discovery runs are not publication evidence. The matrix receipt inventories one normalized watchdog record per cell, binding the configured hard limit and elapsed wall time to the child exit status and exact container ID, name, project, and service observed by the supervisor. Missing, timed-out, or cross-cell records are rejected.

These are not LDBC Benchmark Results.

The evidence home and canonical public presentation are at adversari.al/graph.

10 9. Cypher and GQL

The property graph model so far is a Rust API: builders, traversals, and the store contract. `grust-cypher` adds a query and mutation *language* on top of it — a backend-neutral GQL/Cypher layer — without changing that core.

The language is built as a real pipeline, not ad-hoc string handling: a span-bearing lexer, a recursive-descent parser into a typed AST, and a semantic analysis pass that resolves bindings and kinds. A conformance spine — a `GqlFeature` taxonomy with structured errors — records exactly which constructs are supported, planned, or out of profile, so the surface is auditable rather than aspirational.

10.1 A portable read core

Reads run through a Memory *reference executor* over a graph snapshot: `MATCH` and `OPTIONAL MATCH` (with null padding), multi-hop and variable-length paths, a three-valued `WHERE` expression engine, and `RETURN` with aliases, `DISTINCT`, `ORDER BY`/`SKIP`/`LIMIT`, aggregates with implicit `GROUP BY`, `WITH`, `UNWIND`, and `UNION`. The reference is the definition of correct results.

The read core also composes: `CALL { … }` subqueries execute once per incoming row with the outer bindings visible (correlated import-all scoping) and join their `RETURN` columns back onto the row, and `shortestPath(…)` / `allShortestPaths(…)` find minimal-length simple paths per endpoint pair over a relationship segment. Procedures generalize to table-valued functions: `CALL name(args) [YIELD …]` evaluates its arguments against each incoming row (`tvf.range`, `tvf.keys` join the `db.*` catalog procedures).

Backends that can materialize SQL push the bounded read subset down: a query's `MATCH`/`WHERE` filter lowers into backend SQL (Spark and SQLite dialects), while the `RETURN` projection still runs through the shared reference. Pushed results are therefore identical to the reference *by construc-*

tion, and an embedded-SQLite differential oracle checks reference-vs-pushdown row equality on every change.

Applications that intentionally expose a small read-only surface can use `ReadQueryPolicy`, `validate_read_query`, and `run_bounded_read_query`. This is more than a final `LIMIT`: the parser-backed gate rejects updating and unsafe query shapes, while the in-memory reference executor enforces serialized query, parameter, graph, and output sizes; node and edge counts; cumulative candidate scan and expansion work; cumulative intermediate bytes; result rows; range allocation; cumulative path hops; and a cooperative wall-clock timeout. The intermediate budget accounts cloned bindings, expression and aggregate results, and DISTINCT/GROUP keys before a final `LIMIT`. Scalar and table-valued ranges also keep a library-wide `MAX_RANGE_ITEMS` ceiling. Correlated `CALL { ... }` subqueries charge every repeated node/adjacency index build, and catalog procedures charge each graph scan, so an outer row cannot reset that work or its deadline. Authorization, tenant-safe graph projection, remote-backend deadlines, and process isolation remain the host's responsibility; the cooperative timeout is not an operating-system hard kill.

10.2 Values, procedures, transactions, writes

The value model gains first-class lossless `Decimal` (SQL `DECIMAL`-style) and ISO 8601 `Duration` types alongside the temporal `DateTime`, with parsing, ordering, and checked arithmetic wired through every backend. Read-only catalog procedures (`CALL db.labels()`, `db.relationshipTypes()`, `db.propertyKeys()`) expose schema metadata, and a `START TRANSACTION`/`BEGIN`/`COMMIT`/`ROLLBACK` command surface pairs with honest per-backend atomicity capability reporting. Caller-owned DDL metadata can also be materialized as a portable `CypherCatalogSnapshot`, with deterministic metadata tables for `db.graphs`, `db.graphTypes`, `db.indexes`, and `db.constraints`. Read queries may include `USE <graph>`; the default single-graph executor accepts `USE default`, and callers can bind a graph snapshot to another explicit graph name. Standalone `USE`, `SET`, and `RESET` commands update portable `CypherSession` state without changing transaction-control behavior. Fixed-length path bindings now return first-class `Value::Path` values while preserving the existing JSON path shape, and `Value::Graph` adds first-class set-shaped graph values (deduplicated node/relationship sets built with `graph(nodes, relationships)`). For work that deliberately steps outside the portable surface, `NativeQuery` is an explicit backend-native escape hatch with per-backend language capability flags — FalkorDB accepts native Cypher, SurrealDB accepts SurrealQL, and the SQL backends accept their own dialects — with structured non-support everywhere else.

The writable subset routes acceptance through the same standards-conformant parser, keeping the mutation plans byte-identical to the established write planner while narrowing the public surface to standard Cypher. Pattern-driven writes widen to multiple relationship patterns per statement, incoming `<-[:T]-` edges, and cross-variable correlated `SET`.

10.3 A backed profile

What the layer claims to support is stated precisely in `docs/GQL_PROFILE_STATEMENT.md`: the realized profile is the set of `Supported` features in Grust's scoped manifest. The internal profile is named `Full39075`, but it is not a claim of complete ISO/IEC 39075 certification or uniform backend execution. Sixty-nine of the 74 Grust-catalogued features are implemented; the other five are intentional strict-write rejections. A test pins that scoped-out set to the feature manifest, while backend descriptors and integration tests record which execution paths are reference, pushed, native, or unsupported.

11 10. Example: A Conference Graph

Here is a complete graph-building example:

```
use grust::prelude::*;

fn conference_graph() -> Graph {
    let mut g = Graph::builder();

    let rust = g
        .node("Conference", "conf:rust-graph-day")
        .prop("name", "Rust Graph Day")
        .prop("city", "San Francisco")
        .finish();

    let talk = g
        .node("Talk", "talk:backend-neutral-graphs")
        .prop("title", "Backend-Neutral Graphs in Rust")
        .prop("track", "systems")
        .finish();

    let ada = g
        .node("Person", "person:ada")
        .prop("name", "Ada Example")
        .prop("role", "speaker")
        .finish();

    g.edge("HAS_TALK", &rust, &talk).finish();
    g.edge("PRESENTED_BY", &talk, &ada)
        .prop("confirmed", true)
        .finish();

    g.build()
}
```

The resulting graph can be loaded into memory, exported to CocoIndex target state, or sent to a configured backend:

```
# use grust::prelude::*;
# async fn load<S: GraphStore>(store: &S, graph: Graph) -> grust::Result<()> {
let report = store.put_graph(&graph).await?;
assert_eq!(report.nodes, 3);
assert_eq!(report.edges, 2);
# Ok(())
# }
```

The same graph can be loaded through the typed backend path by declaring the schema and using `put_typed_graph`:

```
# use grust::prelude::*;
# async fn typed_load<S: GraphStore>(store: &S, graph: Graph) ->
grust::Result<()> {
let schema = GraphSchema::builder()
    .node(
        "Conference",
        vec![
            Field::required("name", FieldType::String),
            Field::required("city", FieldType::String),
        ],
    )
    .node(
        "Talk",
        vec![
            Field::required("title", FieldType::String),
            Field::required("track", FieldType::String),
        ],
    )
    .node(
        "Person",
        vec![
            Field::required("name", FieldType::String),
            Field::required("role", FieldType::String),
        ],
    )
    .edge(
        "HAS_TALK",
        vec![Label::new("Conference")],
        vec![Label::new("Talk")],
        Vec::<Field>::new(),
    )
    .edge(
        "PRESENTED_BY",
        vec![Label::new("Talk")],
        vec![Label::new("Person")],
        vec![Field::required("confirmed", FieldType::Bool)],
    )
    .build();

let report = store.put_typed_graph(&schema, &graph).await?;
assert_eq!(report.nodes, 3);
assert_eq!(report.edges, 2);
# Ok(())
# }
```

That call checks node labels, edge labels, required fields, field types, and edge endpoint labels before writing. Backends then use the same schema in their own way: memory validates, LanceDB mirrors into typed Arrow tables, Sail mirrors into typed Delta tables, pgGraph exposes typed SQL views and indexes, SurrealDB defines schemafull tables and fields, and FalkorDB creates useful indexes.

Traversing from a conference to speakers becomes a backend-neutral expression:

```
let speakers = store
    .traverse(
        Traversal::from_node("conf:rust-graph-day")
            .out("HAS_TALK")
            .to("Talk")
            .out("PRESENTED_BY")
            .to("Person"),
    )
    .await?;
```

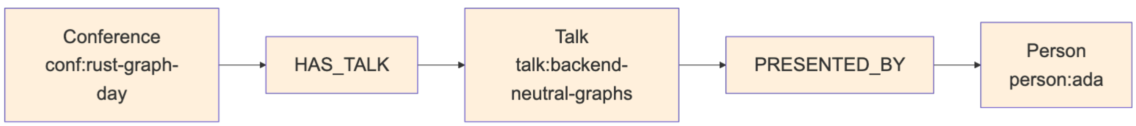

Figure 6: Diagram 6

12 11. Schema and Validation Direction

Grust has schema types: `GraphSchema`, `NodeType`, `EdgeType`, `Field`, `FieldType`, `EdgeUniqueness`, and `GraphConstraint`. `GraphSchema` validates labels, required fields, field value types, edge endpoint labels, edge direction, declared edge uniqueness, and required-property constraints. `FieldType` includes scalar strings, integers, floats, booleans, RFC 3339 date-times, string/int/float arrays, and JSON. Date-time values use the opaque `RfcDate` type internally; construct them through `Value::datetime` or `RfcDate::parse` so invalid strings cannot bypass validation. The default `GraphStore::apply_schema` implementation remains a no-op, which lets schemaless or schema-later backends work without ceremony.

Schema becomes more important for backends that want typed tables, indexes, or label-partitioned layouts. The current schema-capable backends use it in different ways:

- Memory stores the schema and validates local writes.
- FalkorDB creates label/property indexes for declared node types.
- Helix validates schema names that can be lowered through dynamic queries.
- LadybugDB can run in untyped dynamic mode or typed schema-applied mode; typed mode validates writes against the applied schema.
- LanceDB creates typed Arrow tables per node and edge label and mirrors writes into them.
- pgGraph/PostgreSQL/PostgreSQL PGQ exposes typed label views and expression indexes over the universal tables.
- Sail creates typed Delta tables per node and edge label and mirrors writes into them.
- SurrealDB lowers schema into `DEFINE TABLE` and `DEFINE FIELD` statements.

apply_schema is therefore a backend metadata hook, not a portable promise that every future write is enforced by the database. put_typed_graph always validates the whole graph with GraphSchema::validate_graph before applying schema metadata and writing. Callers that need backend-independent guarantees can run the same validation before ordinary put_graph or single-element writes. Memory and LadybugDB enforce the applied schema on subsequent local writes; Sail and LanceDB validate writes before mirroring them into typed tables. FalkorDB, Helix, pgGraph, and SurrealDB currently use schema primarily for indexes, query-shape validation, views, or backend-native definitions. Constraint handling follows the same honest-capability rule. Required-property constraints validate through GraphSchema; unique-property constraints validate inside GraphSchema::validate_graph, and the memory backend reports validate-before-write behavior for them. Memory also supports explicit native constraint application through GraphStore::apply_native_constraint: it stores backend-owned required or unique property constraints, validates them against the current graph before accepting the request, honors if_not_exists, and enforces accepted constraints on later writes without requiring typed GraphSchema metadata. Backends that have not added a read-before-write preflight or native enforcement should continue reporting metadata-only behavior through GraphStore::constraint_capability and unsupported native DDL through native_constraint_capability. grust-cypher also exposes apply_cypher_native_constraints for applying parsed CREATE CONSTRAINT DDL through GraphStore::apply_native_constraint.

A flexible backend can keep universal node and edge tables as the portable interchange surface. A typed backend can add native tables, fields, indexes, or constraints behind the same GraphStore trait.

12.1 Semantic Model Projection

The public facade can also project a versioned semantic model into this same property-graph shape. SemanticModelProjection describes datasets, their fields and physical sources, metrics, and named relationships, bound to a positive model version and SHA-256 source-artifact identity.

semantic_model_graph validates nonempty normalized names, SHA-256 formats, dataset references, and per-scope uniqueness before construction. It uses length-prefixed identity components so punctuation cannot alias a structural separator. Every containment and dataset-relationship edge has an explicit stable ID; therefore two differently named semantic relationships between the same dataset pair survive as distinct edges instead of being collapsed by the ordinary builder's structural deduplication policy. That statement describes the constructed Graph. Persistence keeps both only on backends that support explicit edge IDs; structurally keyed stores collapse edges with the same source, label, and destination.

The output does not introduce a special semantic storage protocol. It is an ordinary Graph containing SemanticModel, SemanticDataset, SemanticField, and SemanticMetric nodes plus containment and RELATES_DATASET edges. It can be replay-compared, queried through the reference engine, or persisted through any backend that supports those ordinary graph operations, subject to that backend's documented multi-edge capability. Artifact-specific adapters remain responsible for parsing a source file and computing the hash supplied to the projection.

The conformance fixture is not synthetic: the test loads the packaged Apache Ossie TPC-DS YAML from upstream commit `ddb19f1b135a61c65603f4823a3526e2fab00cf1`, verifies SHA-256 `bafbdc9d0e304ab22a40592f2b6bdfd45cc399c566533cd71343d33380c0d6e1`, parses the document, and replay-compares the resulting five datasets, 31 fields, five metrics, and four relationships. The published `grust-graph` archive includes Apache Ossie's `NOTICE` and Apache-2.0 text beside that exact fixture. `scripts/verify-package-attribution.sh` inspects the generated crate archive so release packaging fails if any of those three files is absent.

The key architectural point is that schema is metadata about a Grust graph, not a replacement for the graph. Application code can begin with plain graph construction, add schemas when operational needs demand it, and still speak the same store trait.

13 12. Design Tradeoffs

The universal node/edge layout appears in multiple backend plans because it is the easiest way to preserve arbitrary property graphs:

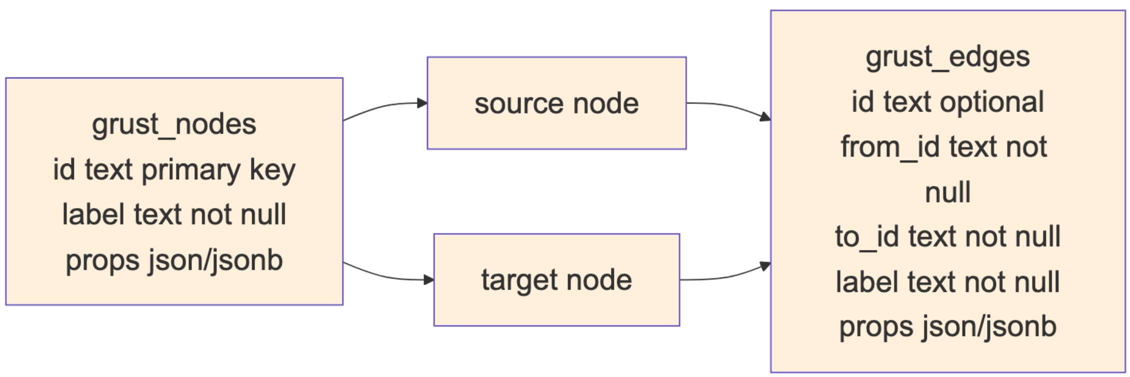

Figure 7: Diagram 7

The cost is that property typing and label-specific optimization require extra work. A label-partitioned backend can use stronger types and better indexes, but it needs schema, migrations, and more planning.

Grust's current architecture keeps both paths open. Core stays universal. Backends choose their storage layout. Schema support can become richer without forcing every backend to look the same.

14 13. Where Grust Can Grow

The next natural step is to deepen graph-native read and traversal support across the backends. HelixDB, LadybugDB, and SurrealDB now satisfy the portable `GraphStore` read and traversal surface, while FalkorDB remains primarily a write and indexing adapter. Further work can push more traversal work into backend-native query forms and add richer result shapes.

Traversal can also grow carefully. Property filters, bounded depth, path returns, shortest paths, and aggregation are all tempting. The important rule is to extend the IR only when several backends can implement the concept without smuggling database-specific query strings through the abstraction.

Incremental mutation uses an extension trait for backends that can apply graph deltas. Its operation model starts with element upsert and deletion and also contains the typed node/edge patch, matched update, property removal, and row-producing operations used by the portable Cypher planner:

```rust
pub enum GraphMutation {
    UpsertNode(Node),
    DeleteNode(NodeId),
    UpsertEdge(Edge),
    DeleteEdge {
        from: NodeId,
        label: Label,
        to: NodeId,
    },
    // Patch and matched-operation variants omitted here.
}
```

That serves CocoIndex-style target-state systems, streaming pipelines, and ordinary applications that need to apply deltas instead of replacing whole graphs. The default `apply_mutations` implementation is ordered but not atomic: if a backend uses the default and a later mutation fails, earlier mutations may already be committed. Backends with real transaction support can override that method. The PostgreSQL and pgGraph stores wrap mutation batches in PostgreSQL transactions, Turso wraps them in a local SQL transaction (which `querygraph-memory` relies on for atomic supersede-and-replace consolidation), and the SurrealDB HTTP and SDK stores wrap mutation batches in SurrealDB transactions. PostgreSQL and Turso's non-returning Cypher plan executors also reject unsupported lowering before writing, then execute the supported operations in source order inside one isolated transaction. The generic write-with-`RETURN` helper remains sequential because later operations may use intermediate bindings; it is not a whole-statement atomicity boundary. Explicit transaction scripts batch supported mutations when atomicity is required.

15 Conclusion

Grust is small by design. Its core model is easy to hold in your head, and its backend contract is narrow enough that very different systems can implement it. That is the source of its leverage.

The project says: build your graph once, keep the domain model in Rust, and let the backend translate. Sometimes that translation is a map scan. Sometimes it is Arrow and LanceDB. Sometimes it is PostgreSQL, Spark SQL, Redis graph commands, SurrealQL, Helix SDK calls, or CocoIndex target state. And sometimes it is a capability-secured agent memory path in which TypeSec guards authority while Grust persists opaque records and their entity graph durably in Turso.

The more Grust grows, the more important that center becomes: a stable property graph model, a backend-neutral traversal IR, explicit errors, feature-gated integrations, and enough Rust type structure to make the right path feel natural.